Roll No. ....

TCE–501

B. TECH. (CE) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

ENVIRONMENTAL ENGINEERING–I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Discuss the various steps taken to prevent theft of water in the distribution system. (CO1)

OR

(b) Describe the peak daily demand and how it affects the design of the water supply system. (CO1)

2. (a) Describe the concept of a coincident draft and how it is used in estimating the design draft for the design of the distribution system. (CO1)

OR

(b) In periods each of 20 years, a city has grown from 60,000 to 1,80,000 and then 2,60,000. Determine the following : (CO1)

(i) the saturation population

(ii) the equation of the logistic curve the expected population after the next 15 years

P. T. O.

3. (a) Give reasons for the following : (CO2)

   (i) the drinking water should possess a high degree of purity.

   (ii) the quantity of water required for industries cannot be connected to the density of the population.

OR

   (b) State how the quantity of water for which a water supply scheme is to be designed is estimated. (CO2)

4. (a) Write short notes on the following : (CO2)

   (i) cement concrete and RCC water mains

   (ii) corrosion of metal pipes

OR

   (b) Discuss briefly the common stress produced in pipelines used for conveying water. What arrangements are made in their design and construction to resist them ? (CO2)

5. (a) Discuss briefly the procedure followed in laying and testing the water supply mains. (CO2)

OR

   (b) A town having a population of 1.5 lakhs is to be supplied with the water from a reservoir 4 kilometres away. It is stipulated that half the daily supply of 225 lpcd will have to be delivered in 8 hours; what should be the size how to supply main if the head available is 16 metres ? Take the value of C for the main as 100 and use the Hazen-William's formula.

(CO2)

Roll No. ....

TCE–502

B. TECH. (CE) (FIFTH SEMESTER) MID SEMESTER EXAMINATION, Oct., 2023

REINFORCED CEMENT CONCRETE–I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain the typical approach of RCC design procedure. Write the advantages and disadvantages of RCC structures. (CO1)

OR

(b) Decide the characteristic strength of the concrete, whose uniaxial compressive strengths values of 40 observations have been supplied as below : (CO1)

29.6	27.3	32.3	38.5	28	33.5	29.5	27.5
31.3	25.5	27	40.1	29.5	35.1	29	26.7
36.7	29.6	32	34	31.3	30.7	32	33.9
30	19.7	21.7	29	31.5	28	33.5	34
30.2	23.8	36.5	30.2	33	23.5	26.9	32

2. (a) Define creep. Discuss the effects and factors-influencing creep. (CO1)

P. T. O.

OR

(b) Determine the weight proportions for M20 nominal mix grade concrete, the properties of the constituent materials are given as below : (CO1)

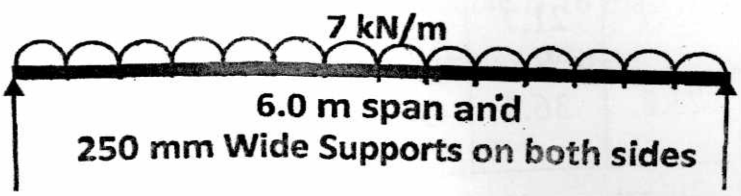
Water-Cement Ration : 0.42%, Void Ratio of Cement, Fine Aggregartes, Coarse Aggregates are 50, 50, 45% and densities are 1500, 1600, 1600 kg/m^3 simultaneously. (CO1)

3. (a) Explain the Stress block parameters of Concrete in Limit state method of design. (CO1)

OR

(b) State and explain the concept of Balanced, Under reinforced, and Over-reinforced flexural sections, while designing the beams and state the key judgements you make as a designer to opt them. (CO2)

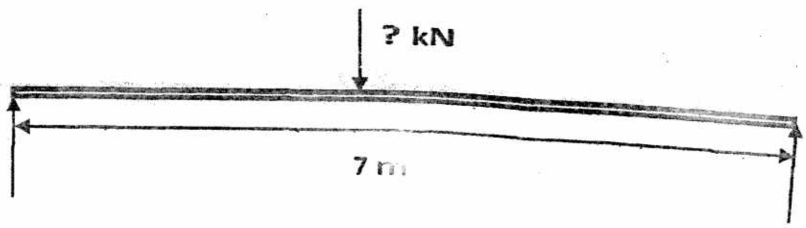
4. (a) Design a rectangular cross-section of the beam in WSM for the maximum bending moment for the given beam in which M20 and Fe 415 grades of material are used. The beam is carrying a UDL of 7 kN/m run throughout the length of the beam. Use 12 mmϕ bars as tension steel. (CO2)

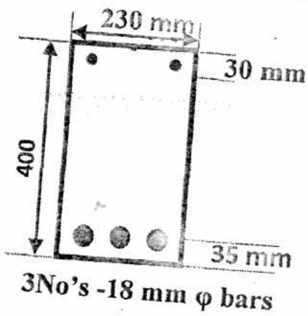


OR

(b) Analyze the given singly reinforced beam section for safety against taking a central point load of 50 kN for the given beam conditions.

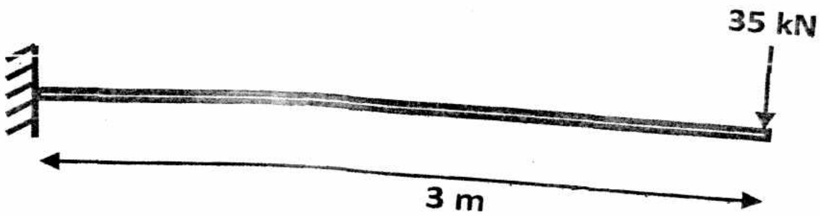
(CO2)





5. (a) Design a rectangular cross section of the beam in WSM for the maximum bending moment for the given beam in which M20 and Fe 415 grades of materials are used. The cross section is restricted to 250 mm × 450 mm. Use 18 mmϕ in compression and 12 mmϕ bars in tension. Assume any other data as per convenience and IS 456 : 2000.

(CO2)



OR

(b) Determine the design Constants (Neutral Axis, Lever Arm, Moment of Resistance and Percentage of Steel Factors) for M20 Grade Concrete and Fe 415 Grade Steel for WSM and LSM of Design.

(CO2)

Roll No. ....

TCE–503

B. TECH. (CIVIL) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

GEOTECHNICAL ENGINEERING-II

Time : Three Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Discuss the step by step procedure for evaluating factor of safety of finite slope using FRICTION CIRCLE METHOD. (CO1)

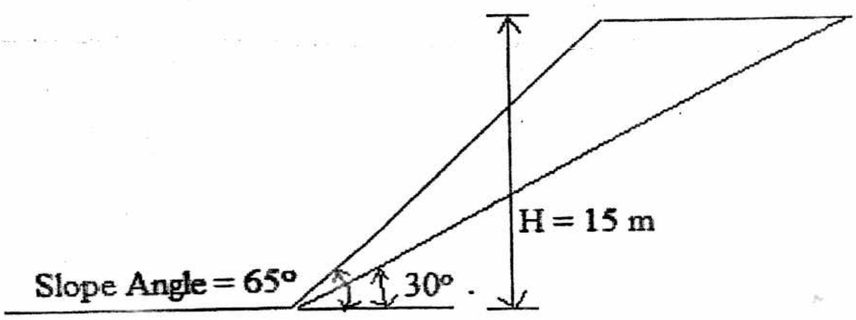
OR

(b) The shearing strength parameters of a soil are : $c'=50\,\text{kPa}$, $\phi'=20°$, $c'_m=30\,\text{kPa}$, $\phi'_m=12°$. Calculate the factor of safety (i) with respect to strength (ii) with respect to cohesion (iii) with respect to friction. The average intergranular pressure $\sigma'=152$ kPa. (CO1)

2. (a) Analyze the factor of safety of the finite slope clay having $c'=32\,\text{kN/m}^2$, $\phi'=26°$, $e=0.68$ and Gs = 2.65. Assume planer failure

P. T. O.

surface as shown in the figure. Use normal graph paper to get information on area of the triangular wedge. (CO1)



OR

(b) Discuss step by step procedure for evaluating factor of safety of the finite slope using "Method of Slices". Develop expression for factor of safety as well. (CO1)

3. (a) Define the following : (CO2)

(i) Ultimate load carrying capacity *vs.* net ultimate load carrying capacity.

(ii) General Shear Failure *vs.* Local Shear Failure.

OR

(b) Define the following terms : (CO2)

(i) Consolidation settlement

(ii) Secondary settlement or creep

(iii) Normally consolidated *vs.* over consolidated soil

(iv) Compression index (Cs)

(v) Coefficient of Compressibility (a_v)

4. (a) List down the assumptions involved in the Terzaghi's bearing capacity equation. Explain how fluctuation in the ground water table affects the load carrying capacity of the foundation soil.

(CO2)

OR

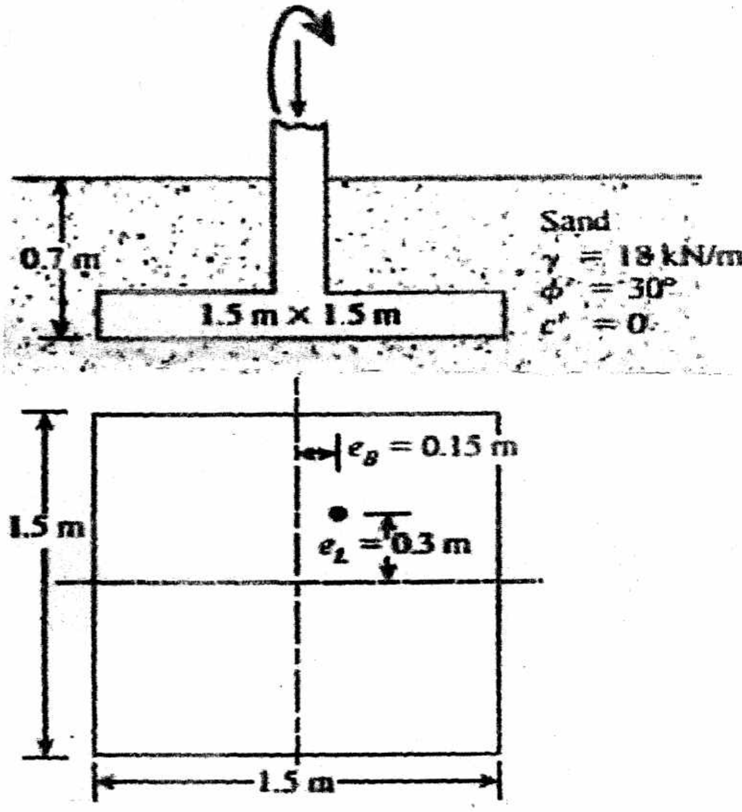
(b) For a shallow foundation : B = 1.5 m, D_f = 1.0 m, Void ratio of the foundation soil (e) = 0.68, Specific gravity of the soil solids (Gs) = 2.65, Estimate the net ultimate load carrying capacity of the foundation using Terzaghi's equation. GWT is at the base of the foundation. Other Geotechnical parameters of the foundation soil are : $c' = 35\,\text{kPa}$, $\phi' = 35°$. Assume general shear failure in the soil and other missing data if necessary.

(CO2)

Bearing capacity factors of Terzaghi

$\phi°$	N_c	N_q	N_γ
0	5.7	1.0	0.0
5	7.3	1.6	0.14
10	9.6	2.7	1.2
15	12.9	4.4	1.8
20	17.7	7.4	5.0
25	25.1	12.7	9.7
30	37.2	22.5	19.7
35	57.8	41.4	42.4
40	95.7	81.3	100.4
45	172.3	173.3	360.0
50	347.5	415.1	1072.8

5. (a) For a shallow strip foundation supported by a silty clay, B = 1.0 m, D_f = 1.5 m, Load per unit are a = 195 kPa and Modulus of Elasticity of the foundation soil is 15×10^6 kPa. Estimate the elastic settlement of the foundation. Give $I_F = 0.787$, $I_E = 0.907$ and $I_G = 0.74$. (CO2)



OR

(b) Estimate the ultimate load carrying capacity of the shallow foundation with two-way eccentricity of the load as indicated in the above diagram. Use Terzaghi's approach. (CO2)

Roll No. ....

TCE–504

B. TECH. (CIVIL ENGINEERING) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

WATER RESOURCES ENGINEERING–I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Define 'hydrology'. Sketch the block diagram showing "an engineering representation of hydrology". What is the need of studying this subject for Civil Engineers ? (CO1)

OR

(b) In a catchment area covering 100 km^2, the average annual precipitation observed at five rain gauge station is as under : (CO1)

Station	Precipitation (mm)
A	750
B	1000
C	900
D	650
E	500

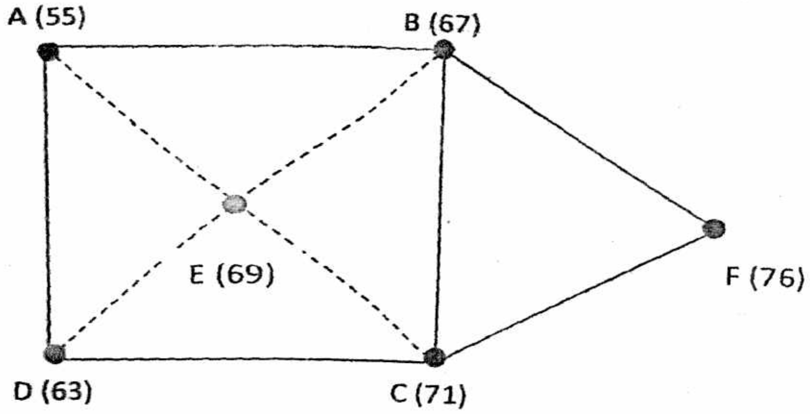
Find the number of additional rain gauge stations and also the rain gauge density if the permissible error is 10%. (CO1)

P. T. O.

2. (a) What factors should be considered in selecting site for raingauge station ? (CO)

OR

(b) Find the mean precipitation for the area sketched below by Theisson method. The area is composed of a square plus an equilateral triangular plot of side 3 kms. Rainfall readings in mm at the various stations are given. Check the same by arithmetic average method. (CO1)



3. (a) Describe the procedure of deriving the ordinates of UH from the given ordinates of flood hydrograph. Discuss the various methods of base flow separation. (CO2)

OR

(b) A basin is divided by 1-hr isochrones in to four areas of size 200, 250, 350 and 170 hactares from the upstream end of the outlet respectively. A rainfall event of 5-hr duration with intensities of 1.7 cm/hr for the first 2 hr & 1.25 cm/hr for the next 3hr occurs uniformly over the basin. Assuming a constant runoff coefficient 0.5, estimate the peak rate of runoff. (CO2)

4. (a) What do you understand by "S-curve Hydrograph" ? Write down the procedure of calculating the ordinate of UH from S-curve. (CO2)

OR

(b) The following data represents the ordinates at hourly interval for one hour unit hydrograph (hydrograph corresponding to one hour duration, effective rainfall of 1 cm). (CO2)

Time (hour)	UHO (cumecs)
0	0
1	580
2	1100
3	960
4	530
5	260
6	140
7	80
8	50
9	40
10	30
11	16
12	10
13	0

Compute the total runoff hydrograph resulting from a two hour storm rainfall with effective rainfall as follows :

First hour = 2 cm

Second hour = 1 cm

Assume a constant base flow of 10 cumecs.

5. (a) The average rainfall over a basin of area of 50 hectares a storm was follows : (CO1/CO2)

Time (hr)	Rainfall (mm)
0	0
1	6
2	11
3	34
4	28
5	12
6	6
7	0

If the volume of runoff from this storm was measured as 25×10^3 m^3. Determine the ϕ-index for the storm.

OR

(b) According to Gumbel's, the estimate flood peaks for a river, based on year of data, for two return periods are : (CO1/CO2)

Return period (years)	Peak flood (m^3/s)
100	485
50	445

Estimate the magnitude of peak flood in river with a return period 1000 years.

Roll No. ....

TCE–505

B. TECH. (CIVIL ENGINEERING) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

STRUCTURAL ANALYSIS—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Determine the horizontal thrust in a semi-circular arch of radius R subjected to a concentrated load W at the crown. (CO1)

OR

(b) Determine the horizontal thrust in a semi-circular arch of radius R subjected to a uniformly distributed load of w per unit length acting over half of the span. (CO1)

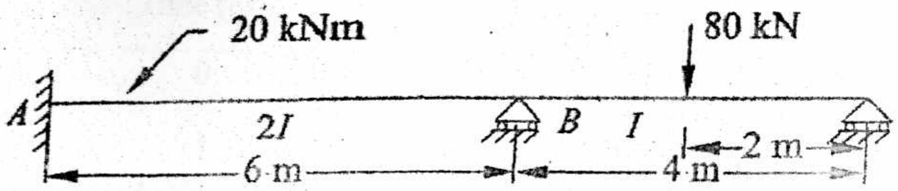
2. (a) A symmetrical two hinged semicircular arch has a span of 20 m and a central rise of 4 m. Find the horizontal thrust at the support if concentrated load of 20 kN acts at the crown. (CO1)

OR

(b) A three hinged segmental arch of span 50 m and central rise of 10 m is subjected to a UDL of 20 kN/m on its left half and a single concentrated load of 500 kN at 12.5 m from the right support. Find the vertical and horizontal reactions at a supports as well as moment at 12.5 from the left support. (CO1)

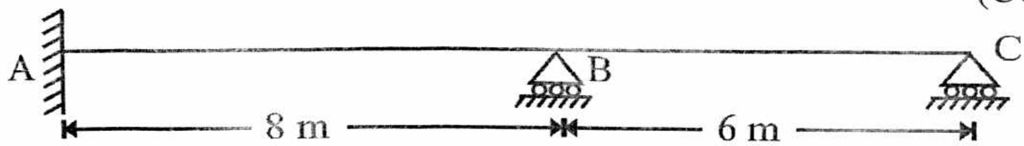
P. T. O.

3. (a) Analyse the continuous beam shown in figure by flexibility matrix method. EI is constant. (CO2)

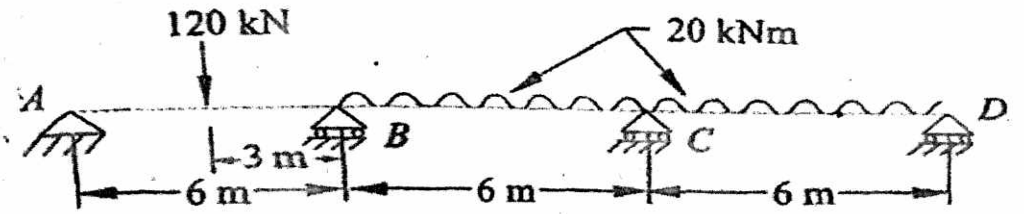


OR

(b) Analyze the continuous beam ABC using Flexibility Method as shown in the figure below, if support B sinks by 10 mm and EI = 6000 × 10^3. (CO2)

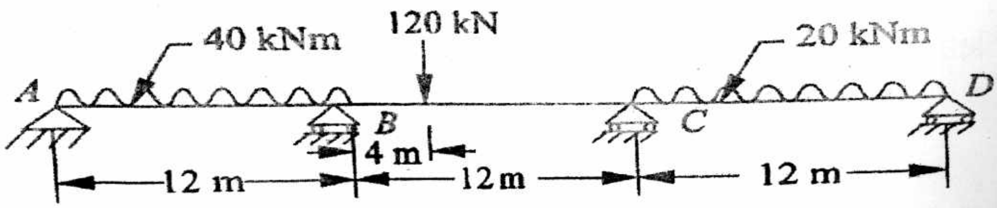


4. (a) Analyze the continuous beam ABCD using Flexibility Method as shown in the figure below : (CO2)

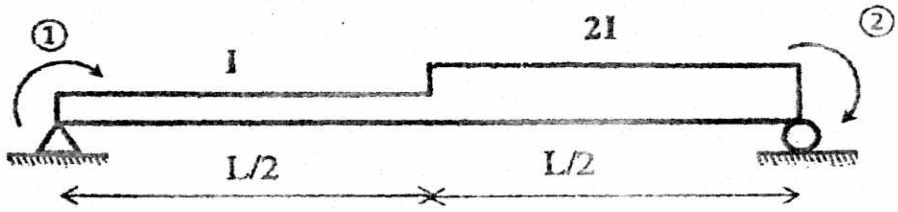


OR

(b) Analyse the continuous beam shown in figure by flexibility matrix method. EI is constant. (CO2)



5. (a) Obtain the Flexibility Matrix of Non-Prismatic Beam shown below : (CO2)



OR

(b) Write short notes on the following : (CO2)

(i) Conditions of Equilibrium

(ii) Degree of Freedom

(iii) Determinate Structures

(iv) Indeterminate Structures.